

WESTERN SYDNEY
UNIVERSITY



Module 4

Logical Operators Extended and Logical Statements

Additional Logical Functions

`find()`

- The `find` function returns the index numbers of the matrix that meets the logical statement given

Type the following into MATLAB:

```
v = [1,3,5,6,3,4;6,36,6,7,843,5];
```

```
[row,col] = find(v == 4)
```

```
row =
```

```
1
```

```
col=
```

```
6
```

Example 1

- The army accepts applicants who are at least 18 old, less than 35 years of age and at least 170 cm tall for the special task force. The following table shows a list of applicants who applied for a position in the special task force.

Height (cm)	Age (years)
180	18
175	19
162	16
165	20
172	17
177	35
173	32

Example 1 (contd.)

- Create a 7 x 2 matrix called applicants which mimic the above table (the columns represent height and age respectively).
- Determine the row numbers that corresponds to the applicants that satisfy the special task force entry requirements.
- Create a 2-column matrix called successful_applicants with entries (rows) that satisfy the entry requirements.

Example 1 - Solution

```
applicants =[180 18; 175 19;162 16;165...  
             20;172  17;177  35;173  32]  
pass = find(applicants(:,1)>=170 &  
applicants(:,2)>=18 & applicants(:,2)<35)  
successful_applicants = applicants(pass,:)
```

Additional Logical Functions

`any()`

- returns a TRUE if **any** of the elements of the vector is a non-zero number or logical 1, otherwise returns FALSE

`all()`

- returns a TRUE if **all** of the elements of the vector is a non-zero number or logical 1, otherwise returns FALSE
 - i.e. if there's one zero element then `all()` returns a FALSE

Using `any()` and `all()`

```
a = randi([0 10],1,3)
```

```
b = randi([0 10],1,3)
```

`all(a>=1)` % to determine if ALL elements in vector a are greater than or equal to 1

`any(a == b)` % to determine if ANY elements between a and b are equal

Other Logical Functions

- **ischar()** returns 1 if () contains character data
- **isinf()** returns 1 if () contains infinity (**Inf**)
- **isnan()** returns 1 if () contains not a number (**NaN**)
- **isnumeric()** returns 1 if () contains numeric data
- **isempty()** returns 1 if () contains empty (**x** = [])

Operator Precedence

- Logical operators are evaluated based on an order/hierarchy (for more detail [click here](#))
- Statements are evaluated from left to right within each precedence level (for example, all **&** and **|** operators are evaluated from left to right).

Operator precedence

1. Arithmetic/PEDMAS/BODMAS
2. All **~** (**NOT**) operators
3. Relational Operators (**<**, **<=**, **>**, **>=**, **~=**, **==**)
4. All **&** (**AND**) operators
5. All **|** (**OR**) operators

Combining relational and logical operations

- To convert the mathematical inequality

$$0 < r < 1$$

into a MATLAB logical expression. It should be coded as

$$(0 < r) \ \& \ (r < 1)$$

or an equivalent statement e.g.

$$(r > 0) \ \& \ (r < 1)$$

- Incorrect values are obtained by typing $0 < r < 1$ (try entering $r = 0.5$)